

PROFILE

Bioinformatics researcher specialising in the intersection of clinical medicine and computational intelligence. I develop high-throughput machine learning pipelines and genomic workflows to solve diagnostic challenges in oncology, neurodegeneration, and longevity science. My work is focused on translating multi-omic data into actionable clinical insights to advance the future of precision medicine.

EDUCATION

MSc Bioinformatics Sept 2025 – Present
University of Birmingham, UK

- Relevant modules: Genomic Data Analysis, Statistical Computing, Machine Learning for Biomedicine, Multi-omics Integration.

MBBS (Bachelor of Medicine & Surgery) 2017 – 2022
J.J.M. Medical College, Davangere, India

- Rotational internship: Medicine, Surgery, Obs/Gynae, Paediatrics, Emergency Care.

Medical Entrance & Refresher Course Aug 2023–Jul 2024
Dr. Bhatia’s Medical Coaching Institute

- Dedicated a period to specialised study and preparation for competitive medical postgraduate examinations.

RESEARCH & PROJECTS

Genomic Pipeline — Thyroid Cancer FNAC-seq 2025 – Present
University of Birmingham

- Building a high-throughput Oxford Nanopore (MinION) long-read sequencing pipeline to resolve diagnostic ambiguity in thyroid fine-needle aspirate cytology.
- Benchmarking against gold-standard histopathology with a view to NHS clinical implementation.

Deep Learning — Medical Image Classification 2024 – Present
Independent Research

- Trained CNNs (TensorFlow/Keras) for multi-class automated diagnostic support; managed full ML lifecycle from data curation and preprocessing to model evaluation.

Predictive Modelling — Dementia & Biological Ageing 2025 – Present
University of Birmingham

- Applied statistical signal processing to longitudinal multi-omic datasets; correlating PhenoAge trajectories with neurological health endpoints to identify early intervention targets.

CLINICAL & RESEARCH EXPERIENCE

Medical Doctor — Government, Private & Nursing Home July 2023 – Jun 2025
Udupi · Davangere · Kochi, India

- Gained frontline primary care experience across government facilities, private clinics, and nursing homes, managing a high-volume, diagnostically varied caseload across both part-time and full-time capacities, interspersed with periods of dedicated postgraduate study preparation.
- Assessed, diagnosed, and managed a broad spectrum of acute and chronic presentations, including diabetes, hypertension, and respiratory conditions; independently initiated and titrated long-term treatment plans.
- Performed minor surgical procedures including wound suturing, IV cannulation, and injections; interpreted ECGs and basic investigations to guide clinical decision-making under resource-limited conditions.
- Within nursing home settings, managed complex elderly and post-operative patients; oversaw long-term polypharmacy medication reviews and led family communication regarding prognosis, discharge planning, and ongoing care needs.

Medical Intern — Rotational Jun 2022 – Jun 2023
J.J.M.M.C. Hospital, India

- Completed structured rotations across Medicine, Surgery, Obs/Gynae, Paediatrics, and Emergency in a high-throughput tertiary hospital setting.
- Actively assisted in general surgical procedures (appendicectomy, hernia repair), obstetric cases (Caesarean sections, episiotomies), and orthopaedic interventions under consultant supervision.
- Managed acute emergency presentations, contributed to ward-round diagnostics, and developed clinical reasoning across diverse inpatient specialties.

TECHNICAL SKILLS

Programming & Data
Python · NumPy · Pandas · Matplotlib · R · Bash · Linux/HPC

AI & Machine Learning
TensorFlow · Keras · Scikit-learn · CNNs · Feature Engineering · Model Evaluation

Bioinformatics Tools
Oxford Nanopore/MinION · GATK · BWA · Samtools · Multi-omics · Variant Calling

Clinical
Diagnosis · Acute Care · History Taking · BLS Certified

CERTIFICATIONS

Deep Learning Specialization
Coursera / DeepLearning.AI · 2025

AI in Medicine Specialization
Coursera · 2025

Basic Life Support (BLS)
Indian Academy of Pediatrics · 2023

RESEARCH INTERESTS
Cancer Genomics · Computational Neuroscience · Neurodegeneration · Longevity & Geroscience · Precision Medicine · Clinical AI · NHS Digital Health

VOLUNTEERING & ENGAGEMENT

Facilitator 2025 – Present
Health Tech AI Hub, UK

- Science communication: translating complex health data for public audiences.

Research Volunteer 2025 – Present
Rejuvenomics, UK

- Longevity biomarker validation bridging molecular geroscience and computation.

Volunteer 2025 – Present
Fit4Surgery Programme, UK

- Patient feedback and peri-operative programme improvement.

Research Assistant — Community Health 2021 – 2022
J.J.M. Medical College, India

- Led data collection and statistical analysis for a rural malnutrition prevalence study; synthesised policy-relevant literature reviews.